

CYBERSECURITY

CARBONEL, NICOLE
CYBERSECURITY NOTES 1

4TH YEAR – PUP BSCPE
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UNDERSTANDING OSI MODELS

What is the OSI model?

- Open Systems Interconnection Reference Model

It is a guide (hence the term "model")

- Avoid focusing excessively on minute details

This is distinct from the OSI protocol suite

- The majority of OSI protocols were not widely adopted

Unique protocols exist at each layer

This model serves as a constant reference throughout a networking career

- It will be referenced frequently

Mnemonic: All People Seem To Need Data Processing

Please - Physical

Do - Data Link

Not - Network

Take - Transport

Susan's - Session

Panties - Presentation

Away - Application

Layer 7 - Application

- The layer we see
- HTTPS, FTP, DNS, POP3
- "Your Eyes"

Layer 6 - Presentation

- Character Encoding
- Application Encryption
- "Application Encryption (SSL/TLS)"

Layer 5 - Session

- Communication management between devices
- Start, stop, restart

- "Control Protocol, Tunneling Protocols"

Layer 4 - Transport

- The "post office" layer
- Parcels and letters
- TCP (Transmission Control Protocol) and UDP (User Datagram Protocol)
- "TCP Segment, UDP Datagram"

Layer 3 - Network

- The "routing" layer
- Internet Protocol (IP)
- Fragments frames to traverse different networks
- "IP Address, Router, Packet"

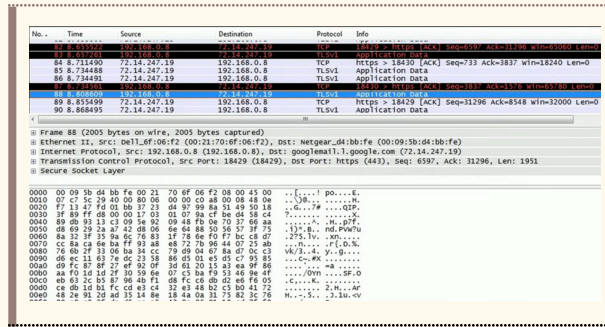
Layer 2 - Data Link

- The basic network "language"
- The foundation of communication at the data link layer
- Data Link Control (DLC) protocols
- MAC (Media Access Control) address on Ethernet
- "Frame, MAC Address, EUI - 48, EUI - 64, Switch"

Layer 1 - Physical

- The physics of the network
- Signaling, cabling, connectors
- This layer isn't about protocols
- "You have a physical layer problem."
- Fix your cabling, punch-downs, etc.
- Run loopback tests, test/replace cables, swap adapter cards
- "Cable, Fibers, and the signal itself"

Wireshark



Summary Protocol

Layer 88

- ★ Application:
https://mail.google.com
- ★ Presentation: SSL encryption
- ★ Session: Link the presentation to the transport
- ★ Transport: TCP encapsulation
- ★ Network: IP encapsulation
- ★ Data Link: Ethernet
- ★ Physical: Electrical signals

Lesson Learned:

In this lesson, I learned about the OSI Model. These are the layers used to transmit data between devices. I have already memorized the layers because they are important for understanding how data is transmitted and which protocols are used at each layer.

NETWORKING DEVICES

Many different ways to forward traffic

- A data center full of equipment
- Every device have a purpose
- The implementation may change over time
- Once installed, It can often be difficult to remove

There are new technologies all the time

- Always something to learn

Router

Routes traffic between IP subnets

- OSI layer 3 device
- Routers inside of switches sometimes called "layer 3 switches"
- Layer 2 = Switch
- Layer 3 = Router

Often connects diverse network types

- LAN, WAN, copper, fiber

Switch

Bridging done in hardware

- Application-specific integrated circuit (ASIC)

An OSI layer 2 device

- Forwards traffic based on data link address

Many ports and features

- The core of an enterprise network
- May provide Power over Ethernet (PoE)

Multilayer Switch

- Includes Layer 3 (routing) functionality

Firewalls

Filter traffic by port number or application

- Traditional vs. NGFW

Encrypt traffic

- VPN between sites

Most firewalls can be layer 3 devices (routers)

- Often sits on the ingress/egress of the network
- Network Address Translation (NAT)
- Dynamic routing

IDS and IPS

Intrusion Detection System /

Intrusion Prevention System

- Watch network traffic

Intrusions

- Exploits against operating systems, applications, etc.
- Buffer overflows, cross-site scripting, other vulnerabilities

Detection vs. Prevention

- Detection – Alarm or alert
- Prevention – Stop it before it gets into the network

Balancing the load

Distribute the load

- Multiple servers
- Invisible to the end-user

Large-scale implementations

- Web server farms, database farms

Fault tolerance

- Server outages have no effect
- Very fast convergence

Load Balancer

Configurable load

- Manage across servers

TCP offload

- Protocol overhead

SSL offload

- Encryption/Decryption

Caching

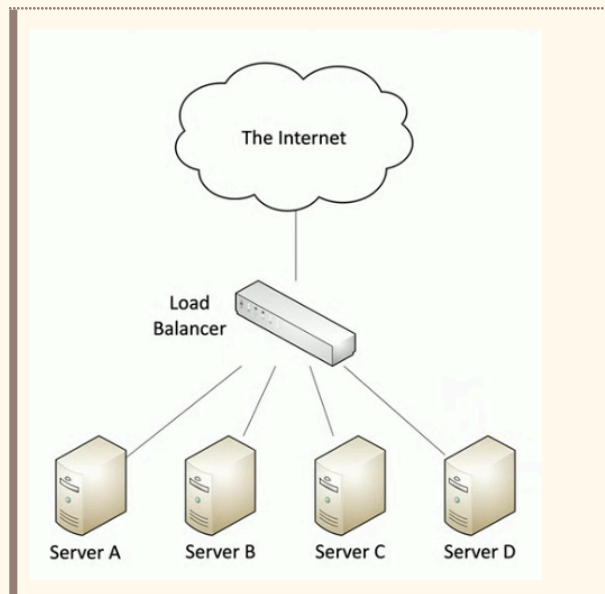
- Fast response

Prioritization

- QoS

Content switching

- Application-centric balancing



Proxies

- Sits between the users and the external network
- Receives the user requests and sends the request on their behalf (the proxy)
- Useful for caching information, access control, URL filtering, content scanning
- Applications may need to know how to use the proxy (explicit)
- Some proxies are invisible (transparent)

NAS vs SAN

Network Attached Storage (NAS)

- Connect to a shared storage device across the network
- File-level access

Storage Area Network (SAN)

- Looks and feels like a local storage device
- Block-level access
- Very efficient reading and writing

Requires a lot of bandwidth

- May use an isolated network and high-speed network technologies

Access Point (AP)

Not a wireless router

- A wireless router is a router and an access point in a single device

An access point is a bridge

- Extends the wired network onto the wireless network
- OSI layer 2 device

Wireless networks everywhere

Wireless networking is pervasive

- And you probably don't just have a single access point

Your access points may not even be in the same building

- One (or more) at every remote site

Configurations may change at any moment

- Access policy, security policies, AP configs

The network should be invisible to your users

- Seamless network access, regardless of role

Wireless LAN Controllers

- Centralized management of access points
 - A single "pane of glass"
- Deploy new access points
- Performance and security monitoring
- Configure and deploy changes to all sites
- Report on access point use
- Usually a proprietary system
 - The wireless controller is paired with the access points

Lesson Learned:

In this lesson, I learned about networking devices that can be used to forward traffic. These are mostly hardware devices that, once installed, can be difficult to remove or replace. I'm already familiar with routers and switches, but I also learned about access points, which I saw during my work immersion at a hospital. These devices are useful for ensuring that a company or building's network operates properly.

NETWORKING FUNCTIONS

There's a lot happening behind the scenes

- Many networking functions are part of the infrastructure

Access to important data

- From anywhere in the world

Remote access

- Secure network communication

Traffic management

- Prioritize the important applications

Protocol support

- Maintain uptime and availability

Content Delivery Network (CDN)

- It takes time to get data from one place to another
- Speed up the process

Geographically distributed caching servers

- Duplicate the data
- Users get the data from a local server

You're using a CDN right now

- Used on many websites
- Invisible to the end user

Virtual Private Network (VPN)

Secure private data traversing a public network

- Encrypted communication on an insecure medium

Concentrator / head-end

- Encryption/decryption access device
- Often integrated into a firewall

Many deployment options

- Specialized cryptographic hardware
- Software-based options available

Often used with client software

- Sometimes built into the OS

Quality of Service (QoS)

- Traffic shaping, packet shaping
- Control by bandwidth usage or data rates
- Set important application to have higher priorities than other apps
- Manage the QoS using router, switches, firewalls, QoS devices

Time to Live (TTL)

How long should data be available?

- Not all systems or protocols are self-regulating
- We sometimes need to tell a system when to stop

Create a timer

- Wait until traversing a number of hops, or wait until a certain amount of time elapses
- Then stop (or drop)

Many different uses

- Drop a packet caught in a loop
- Clear a cache

Routing Loops

Command: `tracert ip address`

Router A thinks the next hop is to Router B

- Router B thinks the next hop is to router A and repeat
- Easy to misconfigure
- This can't go on forever

IP (Internet Protocol)

- Loops could cause a packet to live forever
- Each pass through a router is a hop address
- The router decreases TTL by 1

Frame → Ethernet (src & dst mac address) → IPv4 (src & dst ip address) → DSF, Total Length, Identification → Time to Live (58 hops packets)

DNS (Domain Name System)

DNS Lookups

- Resolve an IP Address from a domain name

A device caches the lookup for a certain amount of time

- How long? TTL second long

Lesson Learned:

In this lesson, I learned that there are many things happening behind the scenes to keep a network running properly. I learned about different networking functions. Some of these concepts were new to me, but I understood that they help with secure communication, managing network traffic, accessing data remotely, and maintaining network availability.

DESIGNING THE CLOUD

- On-demand computing power
- Elasticity
- Application also scale
- Multitenancy

Virtual Networks

- Server farm with 100 individual computers
- All servers are connected with enterprise switches and routers
- Migrate 100 physical servers to one physical servers

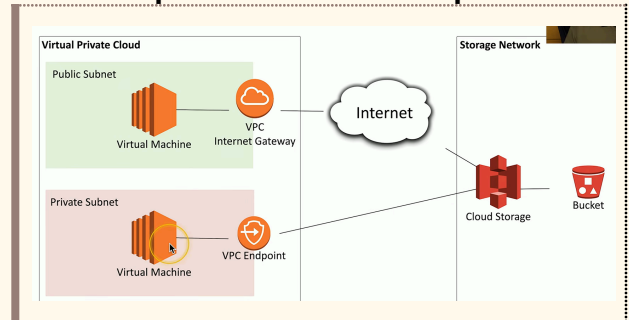
Network function virtualization (NFV)

- Replace physical network devices with virtual versions
- Same functionality as a physical device
- Quickly and easily deploy network functions
- Many different deployment options

Connecting to the cloud

- Virtual Private Cloud (VPC)
- Common to create many VPCs
- Connect VPCs with a transit gateway
- Now make it secure
- VPN (Virtual Private Network)
- Virtual Private Cloud Gateway/Internet gateway
- VPC NAT Gateway
- VPC Endpoint

Virtual private cloud endpoints



Security groups and lists

- A firewall for the cloud
- Layer 4 port number

- TCP or UDP port
- Layer 3 address
 - Individual addresses
 - CIDR block notation
 - IPv4 or IPv6

Network security list

- Assign a security rule to an entire IP subnet
- Applies to all devices in the subnet
- Very broad
 - Can become difficult to manage
- More granularity may be needed
 - Broad rules may not provide the right level of security

Network security group

- Assign a security rule to a specific virtual NIC (VNIC)
 - Applies to a specific devices and network connection
- More granular than network security lists
- Better control and granularity

Lesson Learned:

In this lesson, I learned about the cloud and how virtual networks are used in it. I also learned about the different ways to connect to the cloud and how security works when accessing it.

CLOUD MODELS

- Public Cloud
- Private Cloud
- Hybrid Cloud

Software as a service (SaaS)

- On-demand software
 - No local installation
- Simply log in and use the app

- Central management of data and application
- A complete application offering
 - Google mail, office 365

Infrastructure as a service (IaaS)

- Sometimes called hardware as a service (HaaS)
- You're still responsible for the management
- Your data is out there, but more within your control
- Web server providers

Platform as a service (PaaS)

- No servers, no software, no maintenance team, no HVAC
- You don't have direct control of the data, people, or infrastructure
- Put the building blocks together
- Develop your app from what's available on the platform
 - Salesforce(.)com

Cloud Responsibility Matrix

	SaaS	PaaS	IaaS	On Prem	
Information and Data	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Devices (Mobile and PCs)	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Accounts and Identities	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Identity and Directory Infrastructure	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Applications	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Network Controls	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Operating Systems	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Physical Hosts	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Physical Network	Customer Managed	Customer Managed	Customer Managed	Customer Managed	
Physical Datacenter	Customer Managed	Customer Managed	Customer Managed	Customer Managed	

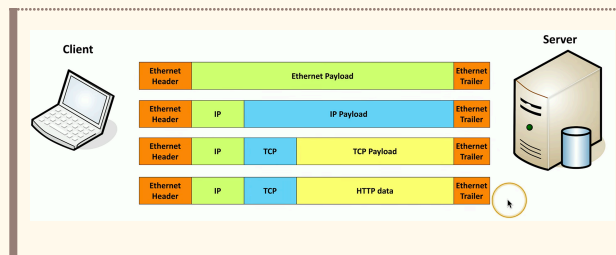
Lesson Learned:

In this lesson, I learned about the cloud models, different kinds of Cloud and the services that the cloud provides. Depending on the type of infrastructure you want and the type of responsibility you need to take on, you need to pick the right cloud model for your application.

Introduction to IP

A series of moving vans

- Efficiently move large amount of data
- The network topology is the road
Ethernet, DSL, cable system
- The truck is the IP
- The boxes hold your data
Boxes of TCP and UDP
- Inside the boxes are more things
Application information
- Client communicating to the server
- Ethernet header → Ethernet Trailer



TCP and UDP

- Transported inside of IP
- Encapsulated by the IP protocol
- Two ways to move data from place to place
- OSI Layer 4
 - Transport Layer
- Multiplexing
- Use many different applications at the same time

TCP - Transmission Control Protocol

- Connection-Oriented
- "Reliable" Delivery
- Flow Control
 - The receiver can manage how much data is sent

UDP - User Datagram Protocol

- Connectionless
- "Unreliable" delivery
- No flow control

- Sender determines the amount of data transmitted

Speedy Delivery

- The IP delivery truck delivers from one (IP) address to another (IP) address
- Boxes arrive at the house / IP address
- Port is written on the outside of the box

Lots of Ports

- IPv4 sockets
- Client / server IP address, protocol, server / client port number
- Non-ephemeral ports - permanent port number
 - ports 0 through 1023
- Ephemeral ports - temporary port number
 - ports 1024 through 65,535

Port Numbers

- TCP and UDP ports can be any number between 0 and 65,535
- Most servers use (services) non-ephemeral (not temporary) port number
- Port numbers are for communication, not security
- service port numbers need to be "well known"
- TCP port number aren't the same as UDP port number ports on the network
- Web server - tcp / 80
- VoIP server - tcp / 5004
- Email server tcp / 143

Lesson Learned:

In this lesson, I learned about the Internet Protocol (IP). It shows how the data is moved from client to server, It also shows the different Protocol that uses in order to communicate the devices properly.